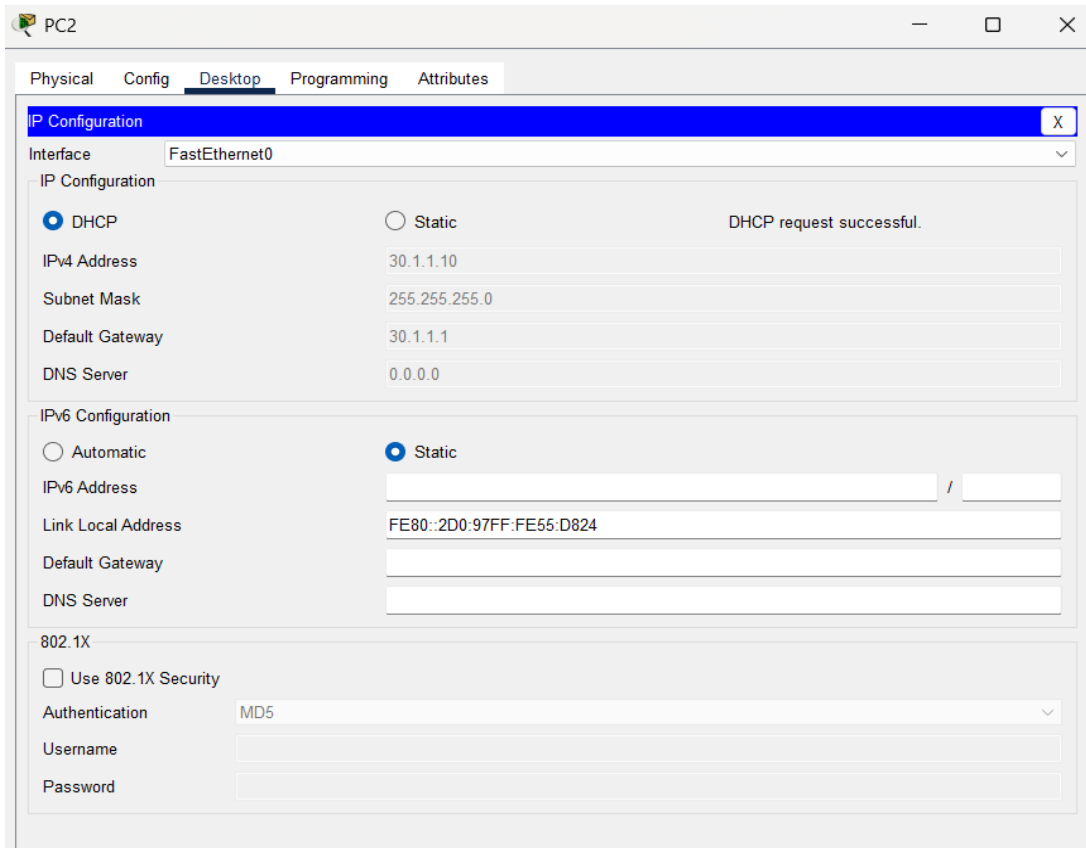


Laboratorio 3.5 — DHCP

Punto 1 — DHCP en la red celeste

Configurado el pool serverPool (Start IP 30.1.1.10, máscara 255.255.255.0, gateway 30.1.1.1), los clientes toman IP automáticamente:



PC2

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 30.1.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 30.1.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2D0:97FF:FE55:D824

Default Gateway:

DNS Server:

802.1X

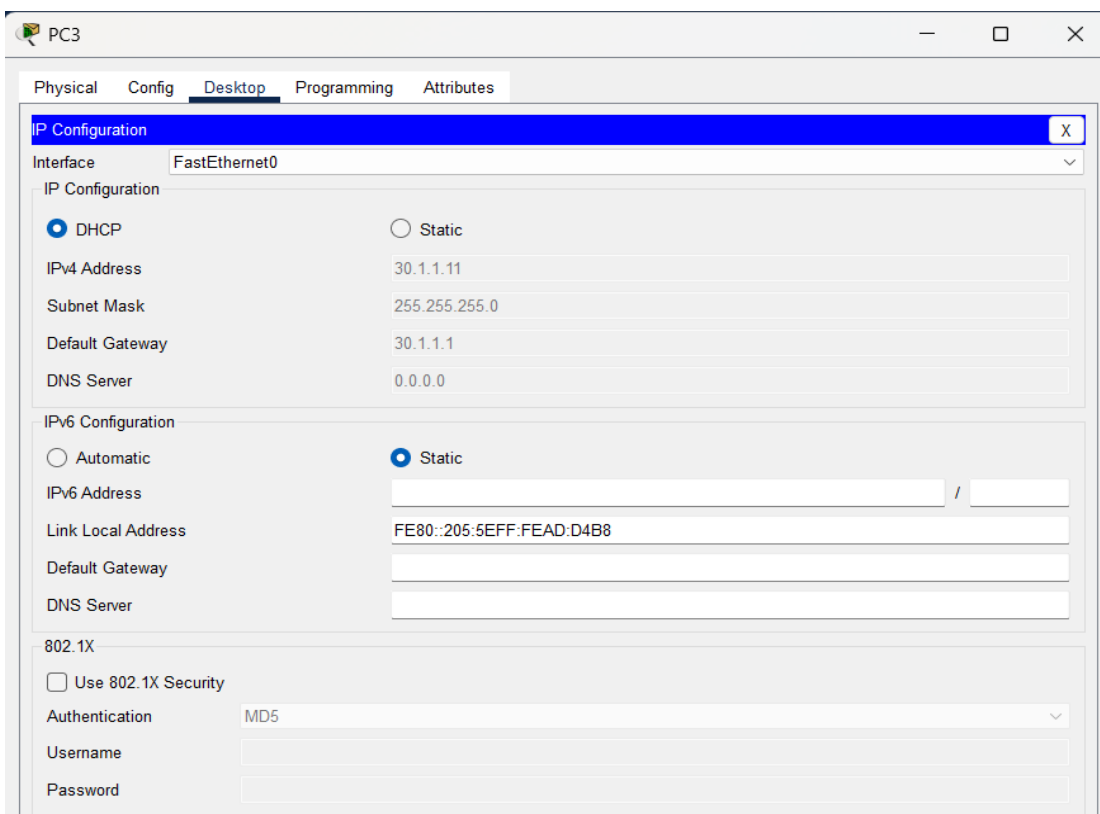
☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

PC2 → 30.1.1.10



PC3

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 30.1.1.11

Subnet Mask: 255.255.255.0

Default Gateway: 30.1.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::205:5EFF:FEAD:D4B8

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

PC3 → 30.1.1.11

The screenshot shows the configuration window for PC10. The 'Desktop' tab is selected. The 'Interface' dropdown is set to 'FastEthernet0'. The 'IP Configuration' section has 'DHCP' selected, with fields for IPv4 Address (30.1.1.12), Subnet Mask (255.255.255.0), Default Gateway (30.1.1.1), and DNS Server (0.0.0.0). The 'IPv6 Configuration' section has 'Static' selected, with fields for IPv6 Address, Link Local Address (FE80::250:FFF:FE27:5A93), Default Gateway, and DNS Server. The '802.1X' section has 'Use 802.1X Security' unchecked, and 'Authentication' set to 'MD5'.

PC10

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 30.1.1.12

Subnet Mask: 255.255.255.0

Default Gateway: 30.1.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::250:FFF:FE27:5A93

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

Cliente nuevo PC10 → 30.1.1.12

Cantidad de mensajes necesarios: 4 (DORA) — DHCPDISCOVER → DHCPOFFER → DHCPREQUEST → DHCPACK.

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.003	Hub2
	0.003	Hub2
	0.003	Hub2
	0.003	Hub2
	1.511	Server0
	1.512	Hub2
	1.512	Hub2
	1.512	Hub2
	1.512	Hub2
	1.513	PC10
	1.514	Hub2
	1.514	Hub2
	1.514	Hub2
	1.514	Hub2
	1.515	Server0
Visible	1.516	Hub2
Visible	1.516	Hub2
Visible	1.516	Hub2
Visible	1.516	Hub2

Reset Simulation
☒ Constant Delay
Captured to: 1.516 s

Play Controls

Event List Filters - Visible Events
DHCP

Edit Filters
Show All/None

Event List
Realtime
Simulation

Los 4 mensajes en el Event List: DISCOVER (PC10) → OFFER (Server0) → REQUEST (PC10) → ACK (Server0).

Punto 2 — Segundo pool (Verdepool)

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: Verdepool

Default Gateway: 20.1.0.1

DNS Server: 0.0.0.0

Start IP Address: 20 1 0 10

Subnet Mask: 255 255 0 0

Maximum Number of Users: 200

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Verdepool	20.1.0.1	0.0.0.0	20.1.0.10	255.255.0.0	200	0.0.0.0	0.0.0.0
serverPool	30.1.1.1	0.0.0.0	30.1.1.10	255.255.2...	240	0.0.0.0	0.0.0.0

Los dos pools configurados: Verdepool (20.1.0.0/16) y serverPool (30.1.1.0/24).

PC5

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP failed. APIPA is being used.

IPv4 Address: 169.254.97.14

Subnet Mask: 255.255.0.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:F9FF:FE57:610E

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

PC5 no obtiene IP: APIPA 169.254.97.14.

A) ¿Qué ocurre al querer tomar un IP y por qué? Los clientes de la red verde no obtienen IP. El DHCPDISCOVER es broadcast y los routers no reenvían broadcasts entre redes, por lo que el pedido nunca llega al servidor (que está en la red celeste).

B) Posibles soluciones: (1) configurar un relay DHCP en el router con ip helper-address apuntando al servidor; (2) usar un servidor DHCP dedicado en la red verde.

Punto 3 — Relay DHCP en el router (ip helper-address)

Router(config)# interface FastEthernet0/0

Router(config-if)# ip helper-address 30.1.1.2

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface **FastEthernet0**

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 20.1.0.10

Subnet Mask 255.255.0.0

Default Gateway 20.1.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:96FF:FED3:70A

Default Gateway

DNS Server

802.1X

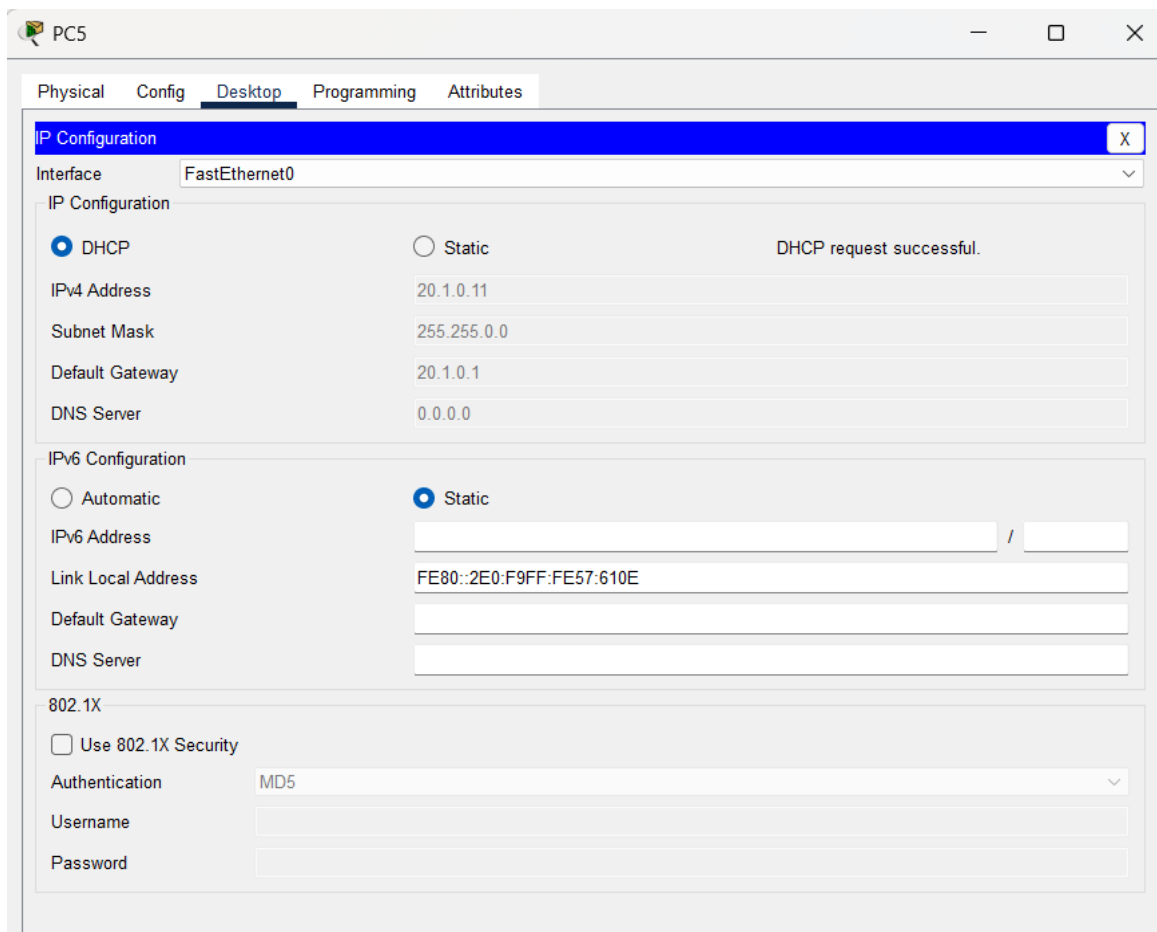
☐ Use 802.1X Security

Authentication MD5

Username

Password

PC4 → 20.1.0.10 (máscara 255.255.0.0) vía relay.



PC5 → 20.1.0.11 (máscara 255.255.0.0) vía relay.

A) ¿Son necesarios los 4 mensajes de DHCP para configurar un host? Sí, los 4 mensajes DORA (Discover, Offer, Request, Ack).

B) ¿Broadcast, unicast o ambos? (un DHCPREQUEST) Ambos, según el tramo:

- Cliente → router (red verde): broadcast (255.255.255.255).
- Router → servidor (relay): unicast al servidor (30.1.1.2), con la IP del router como Relay Agent Address (giaddr = 20.1.0.1).